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Subdaily $PM_{2.5}$ exposure and cardiorespiratory risks: data and findings from Southern California, 2018–2020

Andrew Nguyen[1] · Keita Ebisu[1] · Rupa Basu[1] · Nico Schulte[2] · Scott A. Epstein[2] · Xiangmei Wu[1]

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Abstract

As hourly $PM_{2.5}$ measurements become more accessible, health impacts from subdaily exposures can be evaluated to develop health guidance. We obtained hourly $PM_{2.5}$ concentrations covering Southern California from May 2018 through March 2020 and daily emergency department visits (EDVs) for cardiorespiratory-related conditions at ZIP Code Tabulation Area (ZCTA) levels. ZCTAs were aggregated into 35 clusters based on similar geographic and sociodemographic features. Daily exceedance concentration hours (DECH) above 9, 12, and 15 μg/m^3, daily maximum, and average $PM_{2.5}$ concentrations were calculated for each cluster-day. Two-stage time-series analyses were conducted to estimate excess risks of daily EDVs. DECH metrics exhibited the same direction but smaller effects on cardiovascular and respiratory EDVs compared to daily average metrics. Excess risks for cardiovascular EDVs were 1.77% (95% CI: 1.20, 2.34), 1.04% (0.61, 1.47), and 2.67% (1.98, 3.37) per interquartile range increase of DECH-9, DECH-12, and daily average $PM_{2.5}$ during 7-day lag period, respectively. Excess risks of respiratory EDVs increased by 6.34% (4.25, 8.48), 4.39% (2.85, 5.95), and 6.61% (4.78, 8.47) per IQR increase of DECH-9, DECH-12, and daily average $PM_{2.5}$ during a 3-day lag period, respectively. Elevated excess risks were observed among older adults (65+), children (0–17), and low-poverty neighborhoods on both subdaily and daily metrics. In summary, subdaily $PM_{2.5}$ exposures above the current standards exhibited excess risks in cardiorespiratory-related EDVs but no greater than those derived from the daily average metric. Health guidance based on the daily average metric provides sensible protection to the public in Southern California.

Keywords $PM_{2.5}$ · Hourly concentration · Daily exceedance concentration hours (DECH) · Cardiovascular · Respiratory

Introduction

Ambient exposure to particulate matter with a diameter of 2.5 microns or less ($PM_{2.5}$) has been associated with adverse respiratory, cardiovascular, and metabolic health outcomes (Atkinson et al. 2014; Li et al. 2016; Shah et al. 2015; USEPA 2019). Monitoring of ambient $PM_{2.5}$ with hourly and finer time resolution has become more common, enabling the ability to examine exposures within a day, i.e., subdaily exposure. Subdaily exposure metrics capture exposure in a single hour, exposure during commute hours, daytime average exposure, or the number of hours exceeding a certain limit, so that the peak exposure from short-term events, e.g., fires, industrial operations, fireworks, etc., can be examined. Subdaily exposure may unveil adverse health outcomes that are not observed or that may be attenuated when only measuring at the daily level. However, research on subdaily exposure to $PM_{2.5}$ is sparse, especially in the United States, and the current ambient air quality standards are based on daily and/or annual concentrations. Interpretation of subdaily data and their practical applications to public health remain uncertain.

Globally, literature investigating subdaily exposure to $PM_{2.5}$ has suggested associations with various health outcomes, such as, asthma, chronic obstructive

✉ Xiangmei Wu
xiangmei.wu@oehha.ca.gov

1 Air and Climate Epidemiology Section, Community and Environmental Epidemiology Research Branch, California Office of Environmental Health Hazard Assessment, Oakland, CA, United States of America

2 Planning, Rule Development, and Implementation, South Coast Air Quality Management District, Diamond Bar, CA, United States of America

pulmonary disease (COPD), high blood pressure, acute myocardial infarction (MI), and even hampered cognitive performance (Chen et al. 2020; Cheng et al. 2021; Cleland et al. 2022; Li et al. 2019; Lin et al. 2017, 2021; Zhang et al. 2021). For example, studies in Australia demonstrated elevated risk of MI and asthma aggravation in children within several short (1-hour) to moderate (7–12 h) exposure windows of $PM_{2.5}$ (Cheng et al. 2021, 2022). However, very few of such studies were conducted in the U.S.

Furthermore, it is difficult to consolidate these findings, given the wide variety of subdaily exposure metrics used and various classifications of health outcomes. For example, subdaily metrics used in the literature ranged from single-hour concentrations, cumulative hour windows (e.g., 1–2, 2–6, 7–12 h prior), to aggregate metrics such as daily concentration exceedance hours (DECH). The health outcomes varied from subclinical changes in biomarkers, medication use, outpatient clinic visits, urgent care use, ambulance dispatch, emergency department visits (EDVs), hospitalizations, to mortality. Therefore, the evidence for each individual health outcome is limited and insufficient to build concentration-response relationships for subdaily exposures. Based on the existing limited literature, the United States Environmental Protection Agency's (USEPA) 2019 Integrated Science Assessment (ISA) for Particulate Matter (PM) concluded that subdaily exposures to $PM_{2.5}$ were similar in effect when compared to daily average exposure, with inconsistency across metrics (Jiao et al. 2020; USEPA 2019; Zhang et al. 2021).

Previous research has also suggested differential health burdens of $PM_{2.5}$ across various socioeconomic standing and race/ethnicity groups. Some studies have observed stronger associations between $PM_{2.5}$ and cardiorespiratory outcomes among people of various age groups living in lower socioeconomic status (SES) neighborhoods, while others have reported no modification by neighborhood SES (Alexeeff et al. 2023; Cakmak et al. 2016; Chi et al. 2016; Goodman et al. 2017; Wang et al. 2017). These relationships have not been extensively explored with subdaily exposure metrics (Mehta et al. 2021).

To fill these knowledge gaps, we investigated associations of subdaily exposure to $PM_{2.5}$ and cardiorespiratory EDVs in Southern California to evaluate if subdaily metrics of $PM_{2.5}$ exposure have similar health effects to daily metrics. We also evaluated if there are heterogeneous risks across sociodemographic subgroups by age group, race/ethnicity, and neighborhood poverty level, to identify disproportionate impact across populations.

Methods

Air pollutants

We procured hourly estimations of $PM_{2.5}$ (μg/m^3), ozone (O_3, ppb), and nitrogen dioxide (NO_2, ppb) concentration for the period May 2018 - March 2020 from the South Coast Air Quality Management District (South Coast AQMD). The concentration data was generated using the method used by the South Coast AQMD real-time air quality index map (Schulte et al. 2020). This method combines data from multiple sources: (1) measurements from the South Coast AQMD network of regulatory monitors, (2) $PM_{2.5}$ data from hundreds of PurpleAir sensors (PurpleAir 2023), and (3) modeled ozone and $PM_{2.5}$ fields from the National Air Quality Forecast Capability Community Multiscale Air Quality model (CMAQ) issued by the National Oceanic and Atmospheric Administration (NOAA) (Lee et al. 2017). The measurements and model data are combined using a kriging technique that takes into account the uncertainty of each data source. The resulting hourly concentration data were available at approximate 5 km-by-5 km horizontal grid resolution at the surface and covered the South Coast AQMD jurisdiction - a region including the entirety of Orange County, Los Angeles County except for the High Desert, Riverside County except for the Palo Verde Valley region, and the southwestern portion of San Bernardino County. We assigned the hourly concentrations to ZIP Code Tabulation Areas (ZCTAs) based on the grid that corresponded to their population-weighted centroid. Although $PM_{2.5}$ was our main exposure of interest, O_3 and NO_2 were included as co-pollutants in the sensitivity analyses. Their correlations were examined using Spearman correlation analysis, to ensure that they can be included in the co-pollutant model without having the collinearity issue.

In the dataset, there were a small portion of negative values ($PM_{2.5}$: 1.02% of observations; O_3: 0.98%; NO_2: 0.10%), which is typical for datasets containing low measurements with uncertainty. We replaced all values below the 5th percentile of negative values ($PM_{2.5}$: -4.68 μg/m^3; also for O_3: -3.75 ppb and NO_2: -2.19 ppb) with −1 μg/m^3 or ppb. Extremely high $PM_{2.5}$ concentrations (0.001% of the dataset had $PM_{2.5}$ > 500 μg/m^3, max 1000 μg/m^3), which were likely due to short-term weather events (e.g., dust storms) or local sources (e.g., fireworks or wildfires), were retained in the dataset. Concentration data farther than 140 km from a valid regulatory monitor measurement were removed because these data were subject to higher uncertainty. Missing data were minimal ($PM_{2.5}$: 0.09%; O_3: 0.04%; NO_2: 0.05%) on an hourly basis. While there were some missing hourly estimates for New Years Eve/Day of

2019/2020, we kept them in the analyses because data completeness was greater than 80%.

Daily exceedance concentration hours (DECH), a novel measure that sums the amount of $PM_{2.5}$ exposure over a certain hourly threshold, for the entire day, were adopted as primary subdaily metrics in this study (Lin et al. 2018). For example, DECH-12 refers to the sum of the concentration-hours over 12 μg/m^3 for a day: one hour of 50 μg/m^3 would contribute 38 μg/m^3•hour. DECH-9, 12, 15 were calculated from hourly concentration for each geographic unit. The thresholds of 9 and 12 μg/m^3 were selected to correspond to the USEPA's current and past annual standard for $PM_{2.5,}$ respectively (USEPA 2024). The threshold of 15 μg/m^3 was selected based on the daily average air concentration guideline suggested by the World Health Organization (WHO) (WHO 2021). Daily (24-hour) maximum and daily (24-hour) average were also calculated from hourly concentrations.

Health outcomes

Daily individual patient EDV data from May 2018 to March 2020 were obtained from the Department of Health Care Access and Information (HCAI) to match our exposure data, with each record corresponding to an individual patient's hospitalization/EDV, complete with visit date, residential ZCTA, ICD-10-CM (International Classification of Diseases, Tenth Revision, Clinical Modification) code of primary diagnosis, age, and race/ethnicity data. The exact hourly time stamp of each visit was not available, and therefore, EDV counts were aggregated to daily levels. This analysis focused on EDVs that had primary diagnosis ICD-10-CM codes of respiratory (acute and other) causes (J00 – J99), and cardiovascular-related causes (I00 – I99). Daily counts of EDV cases were summed for each of the 465 ZCTAs within the South Coast AQMD jurisdiction.

Covariates

Demographic and socioeconomic data were procured from the American Community Survey (ACS) for 2015–2019 as collected by the U.S. Census Bureau for each ZCTA in the study area (accessed Sept 2022). These data included proportion of Black, Hispanics, and household income less than two times the federal poverty level. We used these three covariates in the regionalization (described below), and poverty level was also used in the stratified analyses.

Increases in apparent temperature has been found to be positively associated with EDVs of many health outcomes (Basu et al. 2012); therefore, it was adjusted in all analyses. Ambient temperature and relative humidity data were obtained from the USEPA through the Air Quality System Data Mart, California Irrigation Management Information System and National Oceanic Atmospheric Administration (NOAA). Each ZCTA was assigned the apparent temperature, a combination of air temperature and relative humidity that approximates humans' perceptions of cold and heat (Basu et al. 2018), for the nearest monitor from its centroid. The three-day rolling average apparent temperature that covered the same day and two days prior were adjusted in the analysis to account for immediate and lagged cold or heat impact; we utilized this time frame to capture the prolonged effects of extreme temperatures based on previous literature (Basu and Samet 2002; Lin et al. 2009; Peng et al. 2008).

Regionalization

Given the sparse daily EDV counts at the ZCTA level, the model stability and convergence were of concern, as well as potentially wide confidence intervals (CI). To increase the population size of the unit while retaining sociodemographic cohesion, we clustered ZCTAs using a multivariable spatial clustering procedure. Clusters were constructed from spatially contiguous ZCTAs (i.e., sharing at least two vertices or points) with similar race/ethnicity composition (proportion of Black and Hispanic residents) and poverty level (percentage of households with income less than two times the federal poverty limit) utilizing the R *spdep* package and SKATER algorithm (AssunÇão et al. 2006). Noncontiguous ZCTAs (e.g., islands, where a GIS polygon was not connected with surrounding ZCTAs) were reassigned manually using a multinomial logistic regression model matching to the nearest clusters. The procedure was parameterized by specifying the number of clusters in the final assignment and by requiring a minimum population requirement in each cluster. We performed this procedure with trials of different cluster count constraints from 10 to 50, in increments of 5, and different minimum population constraints from ≥ 10,000 to ≥ 50,000, in increments of 5,000.

We optimized cluster fit in two ways: minimizing total mean square error (TMSE) as a function of sociodemographic fit and weighted ANOVA F-statistic as a function of $PM_{2.5}$ measures. A minimized TMSE of the sociodemographic variable fit and the maximized ratio of the between-cluster variability to within-cluster variability measured by the ANOVA F-statistic for $PM_{2.5}$ measures were considered as the best model, with the former indicating cluster separation in sociodemographic characteristics and the latter indicating maximal differentiation in $PM_{2.5}$ variability between clusters.

After optimizing cluster fit, we calculated the sum of daily EDV counts, the population-weighted hourly average for each air pollutant metric ($PM_{2.5}$, NO_2, O_3), the population-weighted daily average for apparent temperature, and the population-weighted census covariates for each cluster.

ZCTAs that were not tabulated by the U.S. Census were excluded, because demographic data may not have been collected, the population was too small to contribute EDV counts, and/or the population was too far from air quality monitors to have reliable estimates. Further, ZCTAs including Veteran Affair's healthcare facilities, college campuses, and amusement parks were also excluded because of their skewed demographic characteristics and/or insufficient case counts in the HCAI database ($n = 10$). The final analytical sample consisted of 455 ZCTAs within the South Coast AQMD jurisdiction.

Statistical analysis

Primary model

We employed a two-stage time-series analyses to estimate the excess risk of daily EDVs associated with various $PM_{2.5}$ exposure metrics, including daily maximum $PM_{2.5}$ concentration and DECH-9, -12, -15 as subdaily exposure metrics. Daily average $PM_{2.5}$ was also examined to serve as a reference and for comparison with existing research.

In the first stage, we used a quasi-Poisson regression with linear distributed lags for $PM_{2.5}$ exposures after controlling for temporal trends and other potential covariates in each cluster. Quasi-Poisson was applied to account for over-dispersed outcome data. In the second stage, individual cluster estimates were combined into an overall estimate using random effects meta-analyses with maximum likelihood estimators.

The first-stage model regressed daily counts of EDVs on each $PM_{2.5}$ metric while also including apparent temperature, a holiday indicator variable, the natural log of a cluster's population, and a natural cubic spline to account for long-term temporal trends (Bell et al. 2008). The model is defined as the following form.

$$\begin{aligned}\log\left[E\left(Y_t\right)\right] = \beta_0 + \beta_1 cb\left(PM_{2.5t}\right) + \beta_2 sma_{t-3}(Temp_t) \\ + ns(t, df) + \beta_3 weekend_t + \beta_4 holiday_t \\ + \beta_5 log(pop) + \in\end{aligned}$$

Where Y_t represents counts of respiratory or cardiovascular-related EDVs on day t. β_0 is the intercept, $cb\left(PM_{2.5t}\right)$ is the crossbasis matrix, which is combined matrices of $PM_{2.5}$ and lag time, explaining the pollutant and lag simultaneously, $sma_{t-3}(Temp_t)$ is the rolling average of apparent temperature for the 3 days preceding the EDV outcome, ns is the natural cubic spline to account for long-term trend, $weekend_t$ is representing a binary weekend indicator variable, $holiday_t$ is the holiday indicator variable, *log(pop)* is the natural log of a cluster's population, and the error term $\in$. The holiday indicator variable included New Year's Day, Memorial Day, Independence Day, Labor Day, Thanksgiving, and Christmas Day.

Model deviance residuals over time were used to diagnose model fit and appropriate control for time trends, as was minimizing quasi-Akaike's Information Criterion (qAIC) for all $PM_{2.5}$ metrics and lag lengths. Model parameterization in lag length, degrees of freedom (d.f.) for distributed lags, and d.f. in natural cubic spline for long-term trend were sourced through prior research. Four different distributed lag metrics (1, 3, 7, and 14 days) were considered - using the *dlnm* package in R (Gasparrini 2011). We fitted polynomial function with 1 d.f. for 1-day and 3-day lags, 2 d.f. for 7-day lag, and 4 d.f. for 14-day lag, respectively. Given the evidence of linear relationship between $PM_{2.5}$ and cardiorespiratory outcomes, we maintained a linear relationship in the distributed lag models of $PM_{2.5}$[1]. For the natural cubic spline controlling for long-term trends, qAIC decreased with larger d.f.: we found that 5 d.f. for cardiovascular-related outcomes and 4 d.f. for respiratory outcomes minimized qAIC.

Stratified analyses

We conducted stratified analysis by patient age group (0–17, 18–64, 65+) and race/ethnicity (non-Hispanic White vs. Hispanics) to investigate effect measure modifications. Black and Asian/Pacific Islander patients were not included because of small case numbers for these groups at the cluster-day level. Datasets stratified by the corresponding age or race/ethnicity categories were fitted to the two-stage model described above. Cardiovascular-related outcomes for patients aged 0–17 years were not examined due to insufficient data.

Furthermore, effect measure modification by cluster level poverty was investigated as a proxy for neighborhood SES. We used Jenks/natural breaks to classify the cluster-level "average percentages of reported household income of less than two times the federal poverty level" into three levels (low (<24%), moderate (24-<38%), and high (≥38%), so that each level has approximately equal sized populations. Tests for interaction where we take the ratio of the difference in risk ratios to their standard errors were performed for all stratified analyses to quantify the degree of heterogeneity in estimates (Altman 2003).

Sensitivity analyses

Potential confounding by co-pollutants were examined. For each $PM_{2.5}$ metric examined in our primary model, we